Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Name	Rising Waters (Flood Prediction System)
Maximum Marks	4 Marks

Technical Architecture:

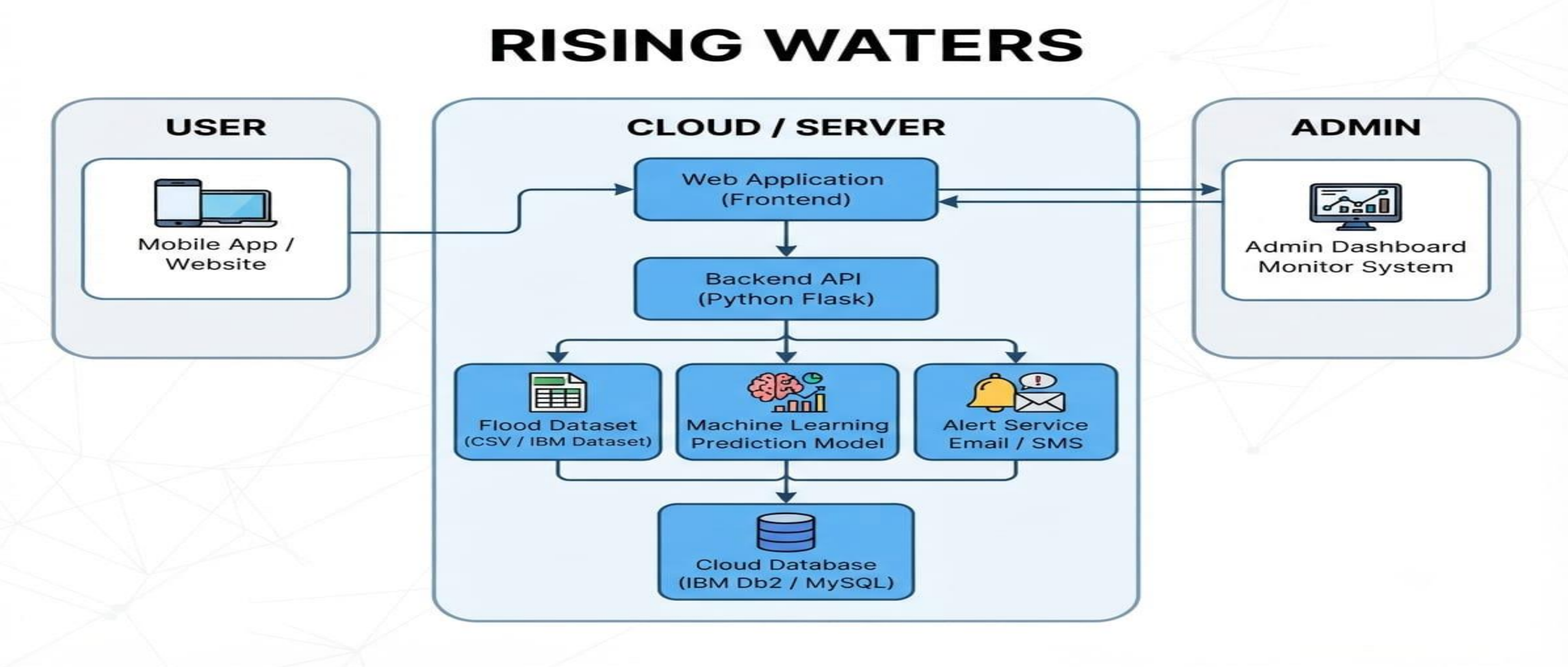

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web page for entering flood parameters and viewing prediction	HTML, CSS, JavaScript
2.	Application Logic-1	Accepts user input	Python(Flask)
3.	Application Logic-2	Processes input data	Python
4.	Application Logic-3	Predicts flood risk using KNN model	Scikit-learn (KNN)
5.	Database	Stores flood dataset.	CSV Dataset
6.	Cloud Database	Not Used	N/A
7.	File Storage	Stores trained model and dataset	Local File System
8.	External API-1	Weather API (optional)	OpenWeather API
9.	External API-2	Not Used	N/A
10.	Machine Learning Model	Flood prediction model	K-Nearest Neighbors (KNN)
11.	Infrastructure (Server / Cloud)	Runs application locally	VS Code, Python, Flask

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used in the project	Flask, Scikit-learn, Pandas, NumPy
2.	Security Implementations	Basic input validation	Flask Security, Input Validation

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Supports future dataset expansion	Flask Architecture
4.	Availability	Accessible whenever server is running	Local Flask Server
5.	Performance	Generates prediction quickly	Scikit-learn KNN